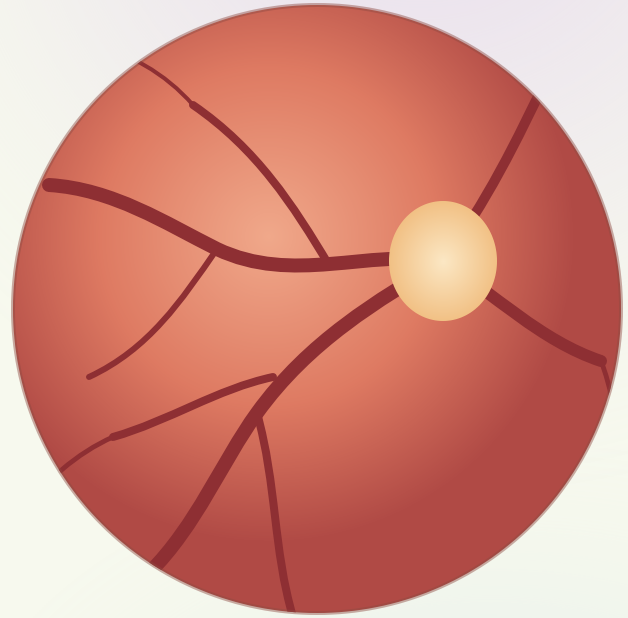
Retinal vessel segmentation

Literature review & base paper selection



Tejmul Movin
sansar.t23csai

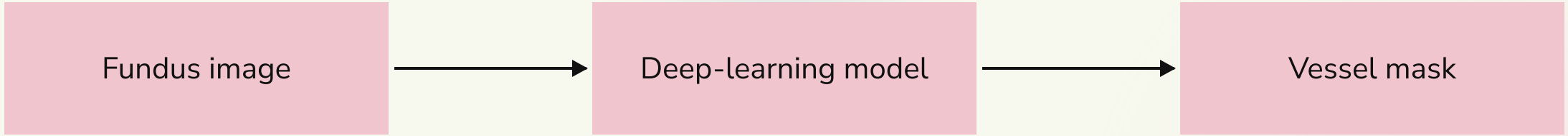
Eranki Sai Vikas
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Can a model trace every vessel?

Binary segmentation: vessel pixel vs. background



Dataset & evaluation

DRIVE — Digital Retinal Images for Vessel Extraction

40
colour fundus photographs

20	20
training	test

565 × 584 pixels · split fixed by the dataset authors · two expert annotators

Metrics

- F1 / Dice
- Sensitivity
- Specificity
- AUC and IoU

Literature Review

Liu et al. (2022), FR-UNet

- Keeps a full-size path through the whole network.
- Nothing is lost to down-sampling, so one-pixel vessels survive.
- Adds a two-threshold step that reconnects weak vessels.
- Limitation: pinned old libraries make it hard to run today.

Dong and Deng (2024), MDFI-Net

- Deformable convolutions inside a pulse-coupled block.
- Keeps faint vessels visible in the deeper layers.
- Reported 97.91–98.16% accuracy over three datasets.
- Limitation: no code was ever released.

Literature Review

Qi et al. (2023), DSCNet

- Sampling points curve along the vessel instead of a square grid.
- Loss charges the model directly for broken shapes.
- Reported 82.06% Dice on DRIVE; ICCV 2023.
- Limitation: changes the convolution and the loss together.

Khan et al. (2023), MRC-Net

- Blends features taken at several scales.
- Trains against a discriminator, rewarding whole regions.
- Published in Neural Networks (Elsevier).
- Limitation: no public code to check the numbers.

Literature Review

Guo et al. (2026), SA-UNetv2

- Attention added at every skip link, not only the bottleneck.
- Loss pairs weighted cross-entropy with a correlation term.
- 82.82% F1 from 0.26M weights; about 1s on a plain CPU.
- Limitation: very new, little independent checking yet.

Liu et al. (2025), MM-UNet

- Mamba-style state-space layers replace pointwise convolutions.
- Aimed at thin, branching shapes that vary across images.
- Reports +1.64% F1 on DRIVE; code released.
- Limitation: the gain is relative, not an absolute score.

Literature Review

Mehmood et al. (2024), LVS-Net

- Light multi-scale encoder with focal modulation at the bottleneck.
- Dice 86.44% using 0.71M parameters, 2.74 MB.
- Built for machines with very little compute.
- Limitation: preprint, and no code was published.

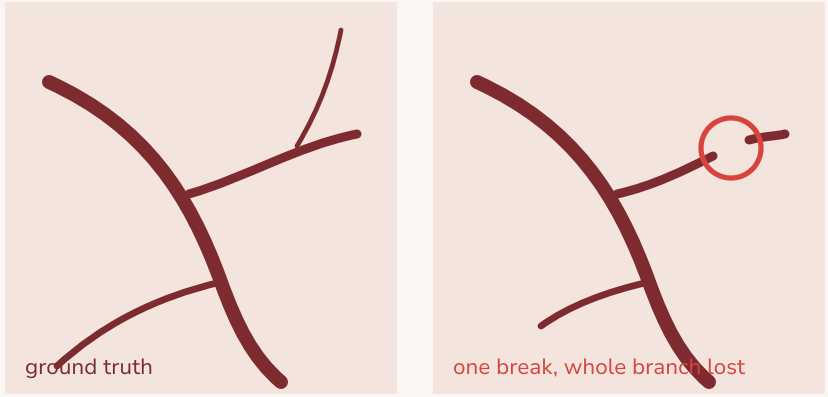
Mehmood et al. (2025), LFRA-Net

- Focal and region-aware attention in selective skip links.
- Dice 84.28% with only 0.17M parameters.
- Code is public on GitHub.
- Limitation: preprint; almost no independent use so far.

Literature Review

Wu et al. (2024), Neuron programming

- Searches over neuron *types* rather than over layouts.
- A small helper network guesses good settings, keeping the search cheap.
- Tested on DRIVE, STARE, CHASE_DB1 and HRF.
- Critical question: does neuron-level search pay off on 20 training images?



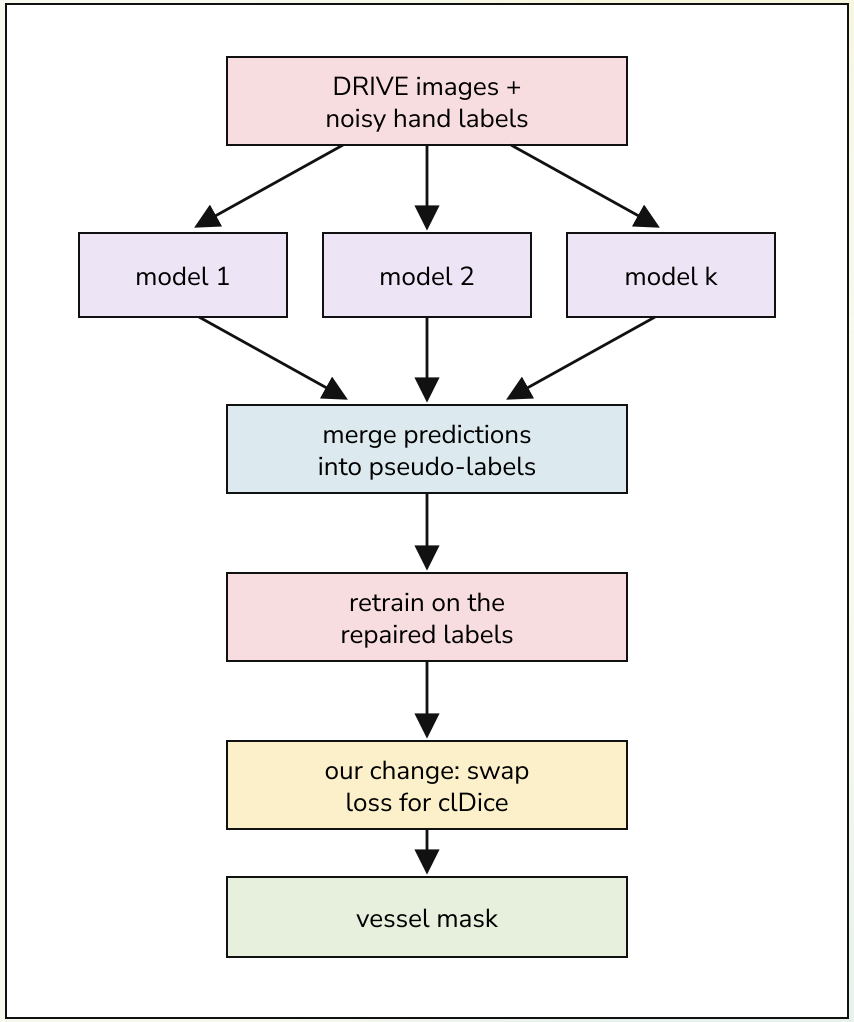
Base Paper

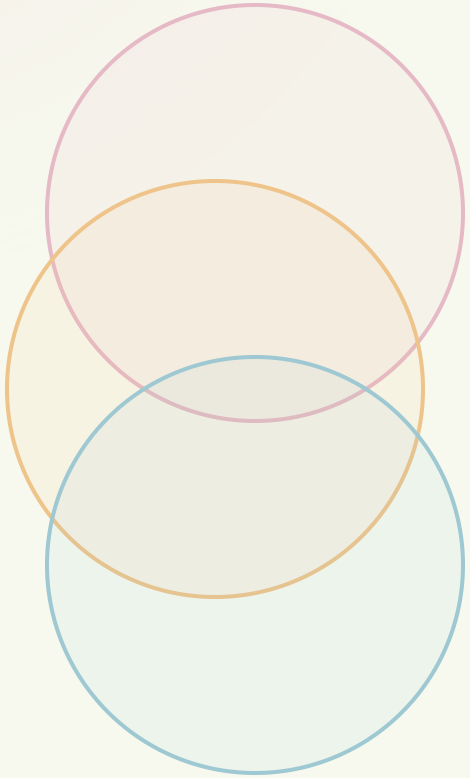
Zhou, Yu and Shi (2021) — SGL

Study group learning: improving retinal vessel segmentation trained with noisy labels · MICCAI 2021

- Argues the ceiling is the *labels*: many thin vessels were never marked.
- Trains k models on different subsets, merges them into cleaner labels, retrains.
- Main strength for us: data, labels and weights all ship in the repository.
- The loss is a command-line flag, so our one change stays one change.

MICCAI	In repo	k = 8	20	1 cmd
2021 venue	DRIVE data	study group	epochs	to verify





Overall findings and next step

- Reported F1 sits between 82.06% and 86.44% — a narrow band.
- Those numbers cannot be compared directly: masking, thresholds and patch sizes all differ.
- Four of the ten papers publish no code, so nobody can check them.
- Every method trains pixel by pixel, and not one reports a connectivity score.
- **Next:** reproduce SGL, then swap its loss for centre-line clDice and measure what moves.

Thank you!

Questions?